

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 1 of 18

(1) Research Program/Project/Study Title: Performance of Lettuce (<i>Lactuca sativa</i> L.) under Hydroponic Culture using Organic Substrate	
Author's Name:	DR. SALVACION J. LEGASPI
Academic Rank:	PROFESSOR VI
Co-Authors (with academic rank):	NOREN MAE E. FELIPE JOEFREY L. LAUNIO
Campus/Satellite College/Address:	CAPSU - BURIAS CAMPUS
Telephone/Cell Phone/Fax Number:	09171162490
E-mail Address:	csrdc@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
Municipality/City:		Mambusao	
Province:		Capiz	
Region:		VI	
(5) Classification Research: ☐ Basic ☒ Applied Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Lettuce		(8) Discipline/Field of Specialization Agriculture	

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 2 of 18

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The study was conducted at the Organic Agriculture Project of CapSU Burias, Mambusao, Capiz, Philippines. It was laid out using the Completely Randomized Design for a two-factor experiment with three replications. The treatments were the different substrates such as A – SNAP, B – CAFSA with chicken dung and molasses, C – CAFSA with cow manure without molasses, D – CAFSA with chicken dung without molasses, and E – CAFSA with hog manure without molasses, and planting schedule which consists of 1st planting – February, and 2nd planting – April. All the data gathered were analyzed using the STAR software, except for the descriptive data. Differences among treatment means were compared using the Least Significant Difference test (LSD) and Duncan Multiple Range Test (DMRT).

The different hydroponic substrates used in the study affected all the parameters except for the stem diameter. SNAP and CAFSA B (with chicken dung and molasses) gave the taller plants, most number of leaves produced, wider and longer leaves, heavier biomass, and heavier marketable part. In terms of the planting schedule, lettuce planted in the first planting that fall in the month of February exhibited a positive result in all the parameters.

A significant interaction among the factors in influencing the width of leaves, length of leaves, weight of marketable part, remaining water, pH, and weight of the sediments were also noted. Among the hydroponic substrates used, lettuce applied with SNAP obtained the highest net income, followed by CAFSA B (with chicken dung and molasses), CAFSA D (with chicken dung and without molasses), and CAFSA C (with cow manure and without molasses).

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Lettuce, Hydroponic Culture, Organic, CAFSA

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Lettuce (*Lactuca sativa* L.) is one of the most popular leafy vegetables consumed worldwide, valued for its nutritional benefits and culinary versatility. The demand for high-quality, pesticide-free produce has driven interest in hydroponic systems, which offer a soilless method of cultivation that can be optimized for various environmental conditions. Hydroponic systems traditionally use inert substrates; however, the incorporation of organic substrates is gaining attention due to their potential to enhance plant growth and contribute to sustainable agriculture practices.

Hydroponic systems provide plants with essential nutrients directly through a water-based solution, allowing precise control over nutrient delivery and leading to faster growth rates compared to traditional soil-based methods (Resh, 2013). Organic substrates in hydroponics, such as coconut coir, compost, and peat, offer additional benefits by improving water retention, providing aeration, and potentially supplying beneficial microorganisms (Diver & Rinehart, 2000). These substrates can enhance the growth and quality of hydroponically grown lettuce by creating a more favorable root environment.

Coconut coir, derived from coconut husks, is a popular organic substrate in hydroponics due to its excellent water-holding capacity, aeration, and biodegradability (Abad et al., 2002). Studies have shown that lettuce grown in coconut coir substrates exhibits improved root development and higher biomass compared to those grown in traditional inert substrates like rockwool or perlite (Siddiqui et al., 2008). The natural properties of coconut

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 3 of 18

coir allow for better root aeration and moisture retention, which are critical for the optimal growth of lettuce in hydroponic systems.

Compost is another organic substrate that can be used in hydroponic systems. Compost not only provides essential nutrients but also enhances the microbial activity in the root zone, which can promote plant health and growth (Pant et al., 2012). Research indicates that compost-based substrates can improve the nutrient uptake and overall performance of lettuce in hydroponic systems, leading to higher yields and better-quality produce (Gravel et al., 2012). The use of compost in hydroponics also aligns with sustainable agricultural practices by recycling organic waste into valuable growing media.

Peat, a natural organic material formed from partially decomposed plant matter, is widely used in horticulture due to its high water retention and nutrient content (Raviv et al., 2004). In hydroponic systems, peat can provide a stable and nutrient-rich environment for lettuce roots, promoting vigorous growth and higher yields (Bunt, 2012). However, concerns about the sustainability of peat extraction have led to increased interest in alternative organic substrates that are more environmentally friendly.

In addition to their physical and chemical properties, organic substrates can influence the microbial communities in the root zone. Beneficial microbes, such as mycorrhizal fungi and plant growth-promoting rhizobacteria, can enhance nutrient uptake, improve plant resilience to stress, and increase overall plant health (Bertrand et al., 2015). The presence of these beneficial microbes in organic substrates can contribute to the improved performance of lettuce in hydroponic systems.

The integration of organic substrates in hydroponic systems also supports the principles of organic agriculture, which emphasize sustainability, environmental health, and the use of natural inputs (Lammerts van Bueren et al., 2011). By using organic substrates, hydroponic systems can reduce the reliance on synthetic inputs, minimize environmental impact, and produce high-quality, organic-certified lettuce (Atkin & Nichols, 2004).

In conclusion, the performance of lettuce under hydroponic culture using organic substrates can be significantly enhanced compared to traditional methods. Organic substrates like coconut coir, compost, and peat offer numerous benefits, including improved water retention, better root aeration, and the provision of beneficial microorganisms. These advantages contribute to higher yields, better quality, and more sustainable production of hydroponically grown lettuce. Continued research and innovation in the use of organic substrates in hydroponics are essential for optimizing lettuce production and promoting sustainable agricultural practices.

Objectives of the Study

The study was conducted to determine the performance of lettuce under hydroponic culture using organic substrates. It specifically aimed to:

1. Determine the performance of lettuce as affected by the hydroponic substrates;
2. Evaluate which among the hydroponic substrates would give the best growth and yield to lettuce;
3. Determine the best planting schedule for growing lettuce using the different hydroponic substrates;
4. Find out if there is an interaction effect between the hydroponic substrates and planting schedule;
5. Determine the profitability of growing lettuce using different hydroponic substrates.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 4 of 18

(11) **Materials and Methods or Methodology** (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The study was conducted at the Organic Agriculture Project of CapSU Burias, Mambusao, Capiz, Philippines. The experiment was laid out using the Completely Randomized Design for a two factor with three replications. The treatments were the different substrates and planting schedule. Different substrates was composed of A – SNAP, B – CAFSA with chicken dung and molasses, C – CAFSA with cow manure and without molasses, D – CAFSA with chicken dung and without molasses, and E – CAFSA with hog manure and without molasses. Planting schedule was consists of 1st planting – February, and 2nd planting – April.

Growth and yield parameters such as plant height, stem diameter, number of leaves, width of leaves, length of leaves, leaf color, biomass, weight of marketable and non-marketable part, pH, volume of the remaining water, weight of sediments, color of the substrate, and % light absorption were gathered and analyzed using the STAR software, except for the descriptive data. Differences among treatment means were compared using the Least Significant Difference test (LSD) and Duncan Multiple Range Test (DMRT).

(12) **Results and Discussions** (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Plant height

Figure 1 presented that the plant height of lettuce. The data shows that plant height was significantly affected with the different hydroponic substrates but not with the planting schedule. There was no interaction effect among factors was observed. Higher plants were recorded when applied with SNAP, while lettuce applied with the different CAFSA showed lower and comparable result.

Stem diameter

The stem diameter of lettuce were not affected with the different hydroponic substrates but resulted a significant difference in the planting schedule. **Figure 2** reveals that bigger stems were observed from the lettuce planted in the first planting than in the 2nd planting.

Number of leaves

The analysis of variance reveals that the number of leaves of lettuce was positively influenced by the different hydroponic substrates and planting schedule. As presented in **Figure 3a**, highest counts were noted from the lettuce applied with the SNAP, followed by CAFSA B (with chicken dung and molasses), CAFSA C (with cow manure and without molasses), and CAFSA D (with chicken dung and without molasses), while the lowest were recorded from the lettuce applied with CAFSA E (with hog manure and without molasses) which was also found similar with the CAFSA C. Additionally, **Figure 3b** shows that lettuce planted in the 2nd planting resulted a highest number of leaf counts than in the 1st planting.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 5 of 18

Width of leaves

The width of leaves of lettuce was highly affected by the two factors. A significant interaction effect among the factors was also noted. **Figure 4** presented that lettuce applied with SNAP in the first planting produced the widest leaves, followed by the lettuce applied with CAFSA B (with chicken dung and molasses) in the 1st planting, and SNAP in the 2nd planting. Narrower leaves were obtained when applied with CAFSA B in the 2nd planting, similar with the lettuce applied with CAFSA C (with cow manure and without molasses), CAFSA D (with chicken dung and without molasses), and CAFSA E (with hog manure and without molasses) in the 1st and 2nd plantings.

Length of leaves

The different substrates and planting schedule affected the length of leaves of lettuce. A significant interaction among factors in influencing the parameter was also observed. The lettuce applied with SNAP and CAFSA B (with chicken dung and molasses) produced significantly longest leaves in the first planting planting. Additionally, lettuce applied with CAFSA B, CAFSA D (with chicken dung and without molasses) and E (with hog manure and without molasses) has a comparable leaf length in the first planting. While the shortest leaves were obtained from the lettuce applied with CAFSA B, C (with cow manure and without molasses), D, and E in the 2nd planting.

Leaf color

The color of the leaves was determined using the leaf color palette for lettuce as shown in **Figure 6**. All of the leaves of lettuce during the first week were similar with the shade B. In the 2nd week, some of the leaf was similar with shade C but a majority of shade B. As the plant grows and the leaf becomes wider, longer and thicker, most of the leaves were similar to shade C especially the plants applied with SNAP and CAFSA B.

Biomass

Biomass of lettuce was highly affected by both factors (**Figures 7a and 7b**). Heavier plants were recorded from those applied with SNAP, followed by CAFSA B (with chicken dung and molasses) which was similar to CAFSA C (with cow manure and without molasses) and D (with chicken dung and without molasses). Lighter lettuce was noticed when applied with CAFSA E (with hog manure and without molasses) which was also found comparable to CAFSA C and D. For the planting schedule, lettuce planted in the month of February was heavier than in the month of April.

Weight of marketable part

The different substrate and planting schedule significantly affected the weight of marketable part of lettuce. There was also a significant interaction within the two factors in affecting the parameter. **Figure 8** reveals that the heaviest marketable part was recorded from the lettuce applied with SNAP in the first planting, followed by CAFSA B (with chicken dung and molasses) in the first planting and SNAP in the 2nd planting. CAFSA C (with cow manure and without molasses) and E (with hog manure and without molasses) during the first planting had similar weights. Lettuce applied with CAFSA B, C, D (with chicken dung and without molasses) and E in the 2nd planting, and CAFSA D during the first planting exhibited a lightest marketable part.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 6 of 18

Weight of non-marketable part

The result of the study revealed that weight of non-marketable part of lettuce were influenced by the difference substrates and planting schedule. **Figure 9a** shows that lettuce applied with CAFSA E (with hog manure and without molasses) exhibited lighter non-marketable part which is comparable to those applied with CAFSA B (with chicken dung and molasses) and C (with cow manure and without molasses). Moreover, the heavier non-marketable were recorded to the lettuce applied with SNAP. For the planting schedule, as shown in **Figure 9b**, lettuce planted in the month of April exhibited lighter non-marketable part.

Survival rate

A 100% survival rates were recorded in the lettuce applied with the different substrates in two planting schedules.

Amount Remaining water

A positive interaction effect among factors was observed on the amount remaining water (**Figure 10**). The effect of the different substrate and planting schedule were also found to be significant. Lettuce planted in the first planting recorded the lowest volume of remaining water when applied with the different substrates. On the second planting, highest volume was noted when applied with CAFSA B (with chicken dung and molasses) and CAFSA E (with hog manure and without molasses), followed by SNAP, CAFSA C (with cow manure and without molasses) and D (with chicken dung and without molasses).

pH of the substrate

The pH of hydroponic substrates affects how well a plant can absorb essential nutrients. Based on the weekly analysis, during the first planting, the pH of the SNAP and CAFSA B (with chicken dung and without molasses) increases every week, while for the CAFSA C (with cow manure and without molasses), CAFSA D (with chicken dung and without molasses), and CAFSA E (with hog manure and without molasses), the pH increased in the 1st and 2nd week and abruptly decreases in the 3rd week.

On the second trial, the pH of SNAP decreased in the 1st and 2nd week and increases in the 3rd week. For the CAFSA B, pH gradually decreased weekly, while the CAFSA D, pH remains constant from the 1st to 2nd week and suddenly decreases on the 3rd week. Finally, the pH of CAFSA C and E increased on the 1st and 2nd week and decreases in the 3rd week.

In the **Figure 11**, the pH was significantly influenced by the interaction within the two factors. A similar and highest pH was noted in the hydroponic substrates applied during the 1st planting which ranges from 7.62-7.79. Additionally, on the 2nd planting, pH of SNAP, CAFSA B and CAFSA C were comparable with a mean if 6.32, 6.20, and 6.23, respectively. The lowest pH was observed in CAFSA D (5.49) and CAFSA E (5.47).

Weight of sediments

The weight of the sediments collected in the substrates during the termination was affected by the two factors. A significant interaction effect was also noted. **Figure 12** reveals that lighter sediments were noted in all substrates during the 1st planting. On the second planting, heavier sediments were recorded in CAFSA B (with chicken dung and molasses), C (with cow manure and without molasses), D (with chicken dung and without molasses), and E (with hog manure and without molasses).

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 7 of 18

Percent (%) light absorption

Both factors had influenced the percent light absorption. As seen in **Figures 14a and 14b**, SNAP absorbed the highest percent of light which was similar with CAFSA E (with hog manure and without molasses), and CAFSA E was also found comparable with CAFSA B (with chicken dung and molasses). Moreover, the lowest % of light was absorbed by the CAFSA D (with chicken dung and without molasses) which was similar with CAFSA C (with cow manure and without molasses).

Using the color wheel palette in **Figure 15**, the colors of the substrates that obtained the highest percent of light absorption were determined. The SNAP solution was colorless, CAFSA E was baby yellow, and CAFSA B were gold ochre. Additionally, the dark colored substrates were observed from CAFSA D (color of brown and red wood) and CAFSA C (color chocolate and chrome orange) which absorbs the lowest percent of light.

Net income over substrate cost of lettuce applied with the different hydroponic substrates

Among the hydroponic substrates used, lettuce applied with SNAP obtained the highest net income amounting to Php 1,313.66, followed by CAFSA B (with chicken dung and molasses), CAFSA D (with chicken dung and without molasses), and CAFSA C (with cow manure and without molasses) with the amount of Php 793.80, Php 600.48, and Php 590.40, respectively. The lowest net income was recorded in the lettuce applied with CAFSA E (with hog manure and without molasses) amounting to Php 531.00. However, after deducting the substrate cost and other production cost as reflected in Table 1, the highest net profit was generated from CAFSA B (P 542.00), followed by CAFSA C (P 355.48). SNAP A generated a net income of P 333.00, CAFSA E has a net income of P286.00 and the lowest net income was generated from CAFSA D (P245.00).

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Based on the results of the study, the following conclusions were drawn:

1. Lettuce varied significantly with the application of the different hydroponic substrates. Application of SNAP and CAFSA B (with chicken dung and molasses) gave the taller plants, most number of leaves produced, wider and longer leaves, heavier biomass, heavier marketable part and highest percent of light absorption.
2. In terms of the planting schedule, lettuce planted in the first planting which fall in the month of February exhibited better result in all the parameters.
3. A significant interaction among the factors in influencing the width of leaves, length of leaves, weight of marketable part, remaining water, pH, and weight of the sediments were also noted.
4. Wider leaves of lettuce were recorded when applied with SNAP in the first planting; it was followed by the application of CAFSA B (with chicken dung and molasses) in 1st planting which was similar to SNAP in the 2nd planting.
5. Longer leaves was noticed in the first trial when applied with SNAP which was comparable with the CAFSA B. A similar length of leaves was also observed in CAFSA B, CAFSA D and CAFSA E in the first trial.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 8 of 18

6. The weight of marketable part is heavier when applied with SNAP, followed by the CAFSA B in the 1st planting which was comparable with the SNAP in the 2nd planting.
7. Higher volume of remaining water was measured in CAFSA B and CAFSA E in the 2nd planting while similar volume of water remained in the 1st planting.
8. A similar and highest pH was noted in the hydroponic substrates applied during the 1st planting while on the 2nd planting, pH of SNAP, CAFSA B and CAFSA C were comparable, and CAFSA D and CAFSA E had the lowest pH.
9. Lighter sediments were noted in all substrates during the 1st planting. On the second trial, heavier sediments were recorded in CAFSA B (with chicken dung and molasses), C (with cow manure and without molasses), D (with chicken dung and without molasses), and E (with hog manure and without molasses).
10. Among the hydroponic substrates used, lettuce applied with CAFSA B (with chicken dung and molasses) generated the highest income.

Recommendations

Based on the result of the study, the following recommendations are forwarded:

1. Use CAFSA B in growing organic lettuce under hydroponic culture for higher net income.
2. Grow lettuce under hydroponic cultivation during the month of April.
3. CAFSA B, CAFSA C, and CAFSA D exhibited a remarkable result and should consider these economic outcomes when selecting substrates, balancing initial investment with potential returns.
4. Conduct similar studies using other crops and planting schedule.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

- Abad, M., Noguera, P., & Bures, S. (2002). National inventory of organic substrates in Spain. *Bioresource Technology*, 77(2), 197-200.
- Atkin, P., & Nichols, M. A. (2004). Organic hydroponics. *Acta Horticulturae*, 648, 121-127.
- Bertrand, G., Nguema-Ona, E., Rippa, S., Sorin, C., Noirot, E., Driouich, A., & Vické-Gibouin, M. (2015). Arabinogalactan proteins and roots of plants: A conserved interface for the interaction with soil and rhizosphere organisms. *Frontiers in Plant Science*, 6, 811.
- Bunt, A. C. (2012). *Media and Mixes for Container-Grown Plants: A Manual on the Preparation and Use of Growing Media for Pot Plants*. Springer Science & Business

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 9 of 18

APPENDICES

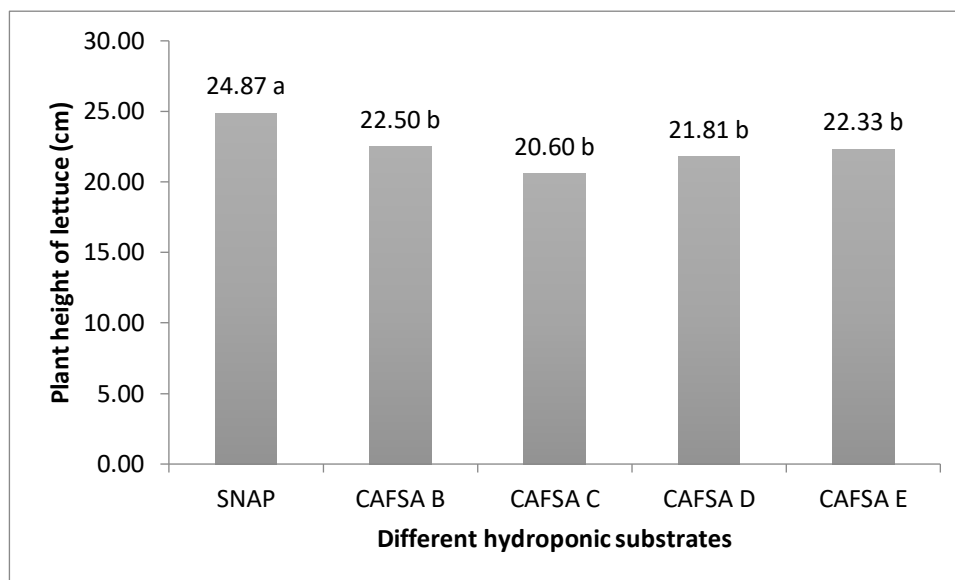


Fig. 1. Plant height (cm) of lettuce as affected with the different hydroponic substrates.

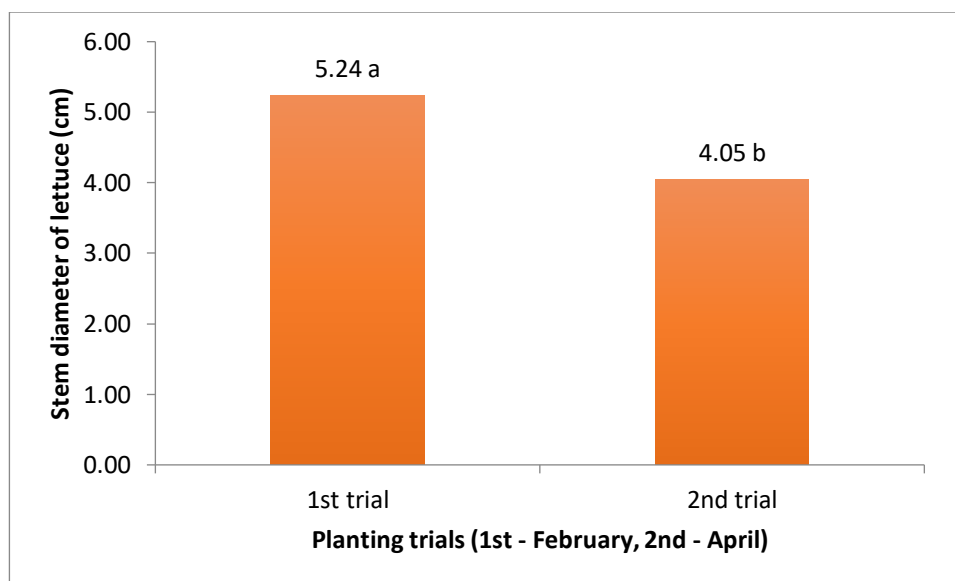


Fig. 2. Stem diameter (cm) of lettuce as affected by the different planting schedule.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 10 of 18

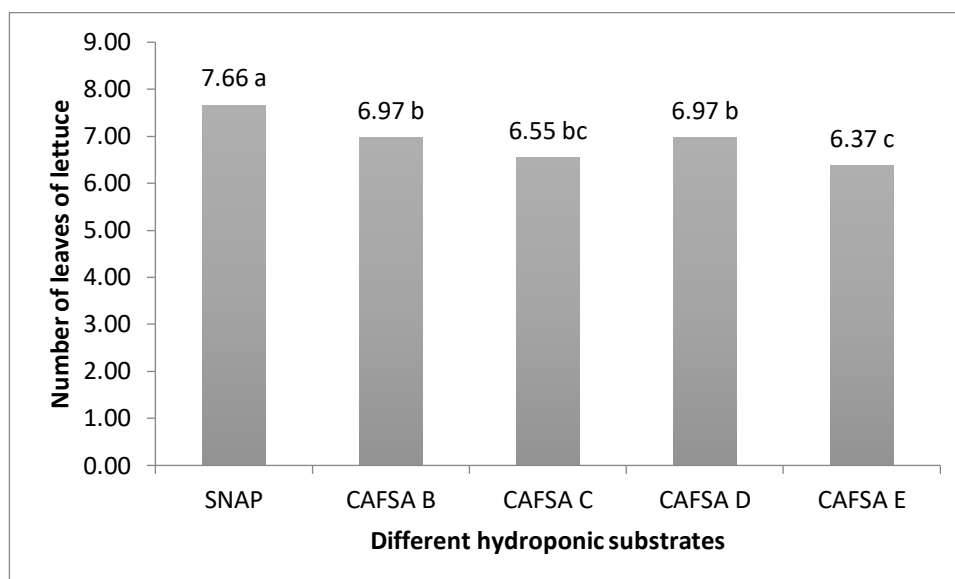


Fig. 3.a. Number of leaves of lettuce as affected with the different hydroponic substrates.

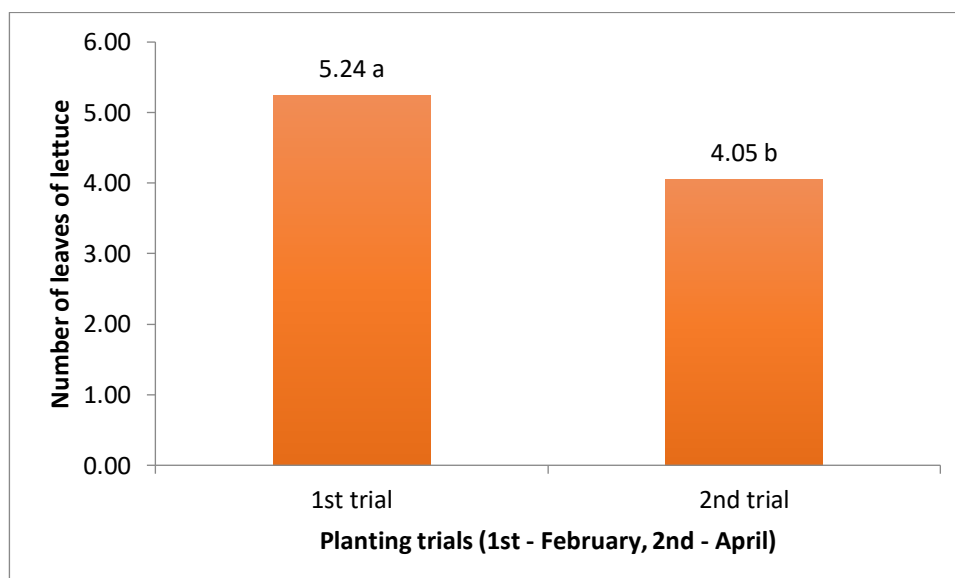


Fig. 3.b. Number of leaves of lettuce as affected with the different planting schedule.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 11 of 18

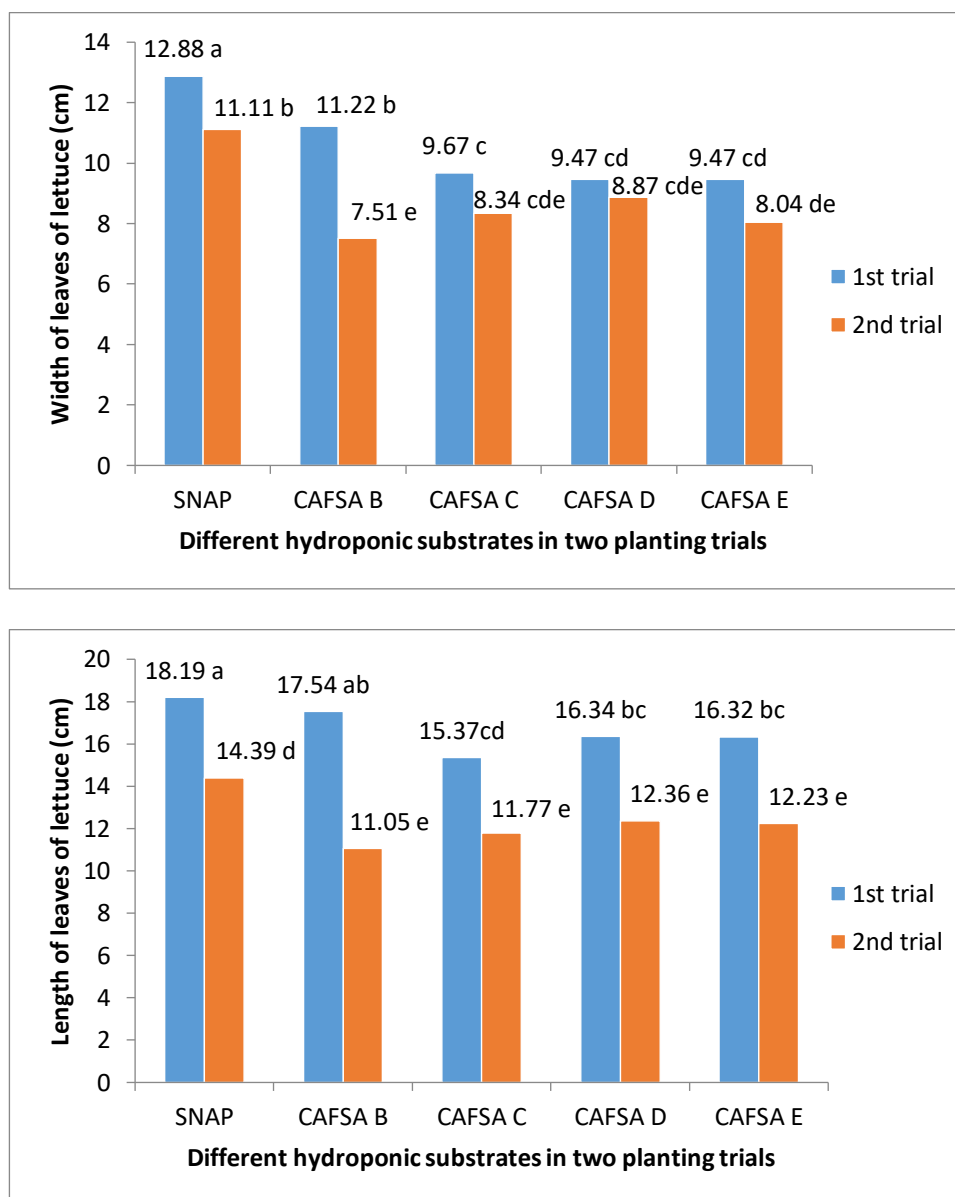


Fig. 5. Length (cm) of leaves of lettuce as affected with the different hydroponic substrates and planting trials.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 12 of 18



Lettuce Palette

Figure 6. Lettuce color palette, accessed at <https://www.dreamstime.com/backgroundcloseup-curly-lettuce-leaves-colour-palette-complimentary-colour-swatches-lettuce-palette-image246679351>

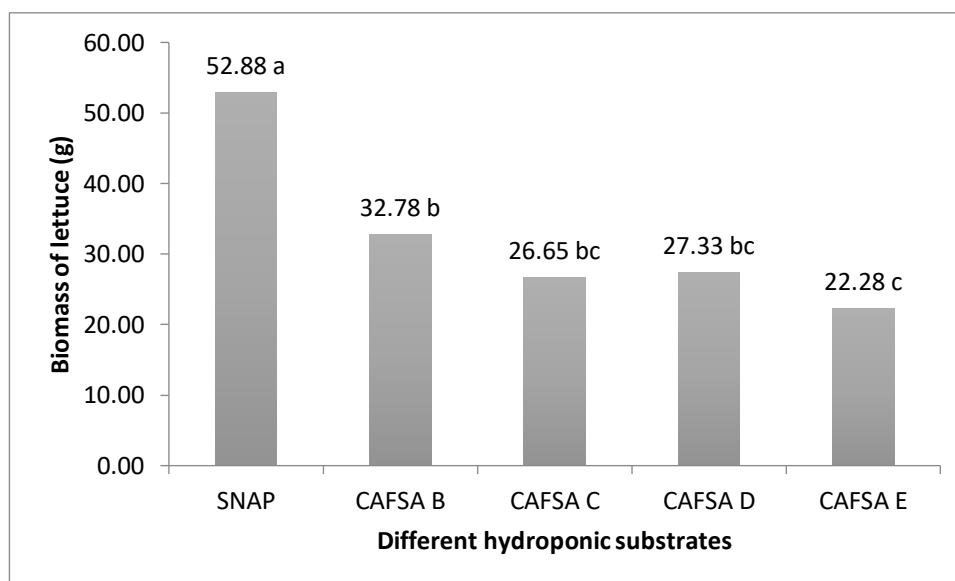


Fig. 7.a. Biomass (g) of lettuce as affected with the different hydroponic substrates.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 13 of 18

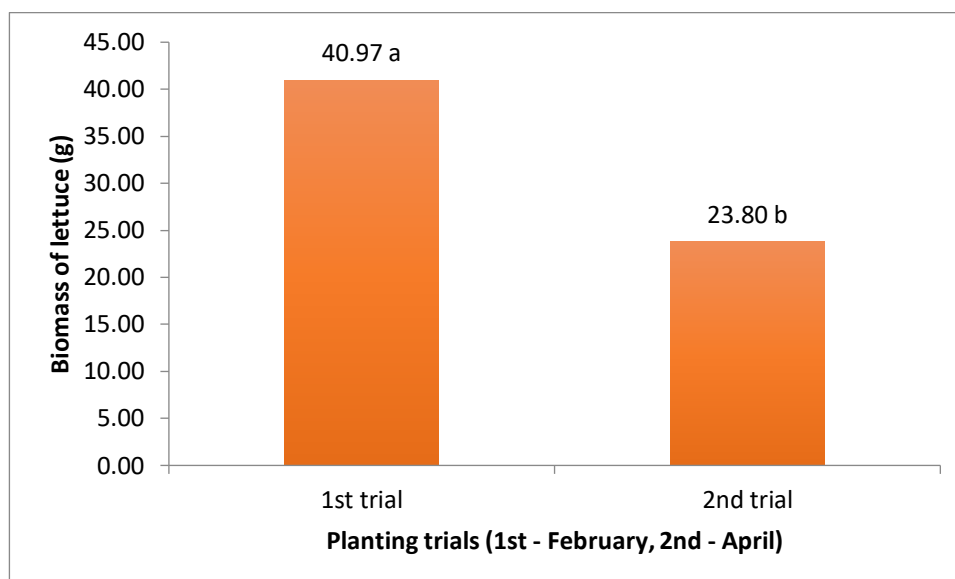


Fig. 7.b. Biomass (g) of lettuce as affected with the different planting trials.

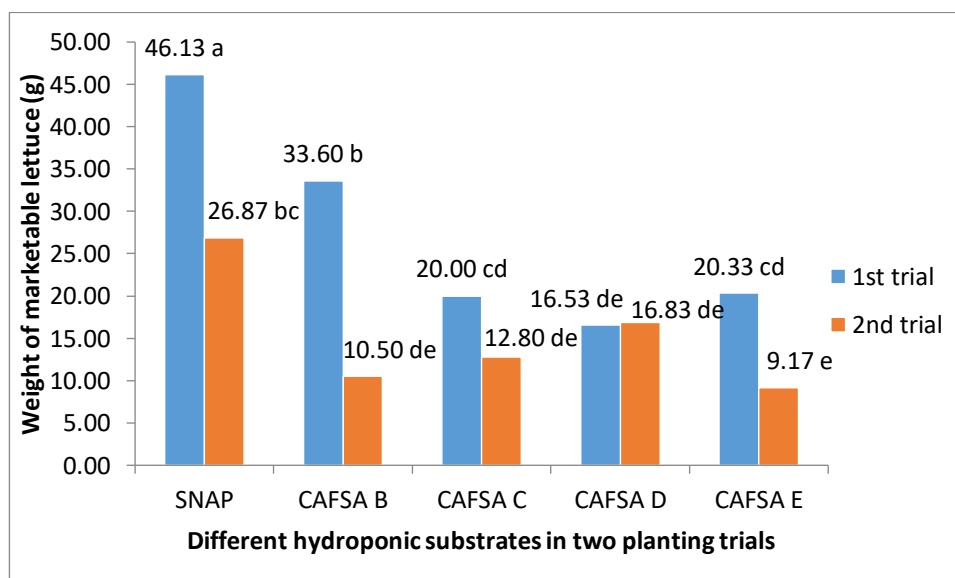


Fig. 8. Weight of marketable lettuce (g) as affected with the different hydroponic substrates and planting schedule.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 14 of 18

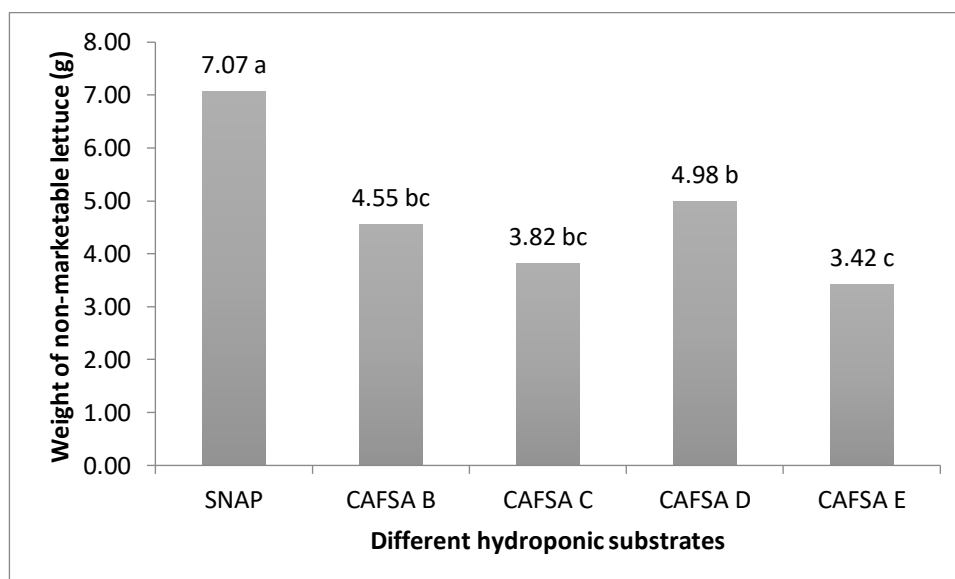


Fig. 9.a. Weight of non-marketable lettuce (g) as affected with the different hydroponic substrates.

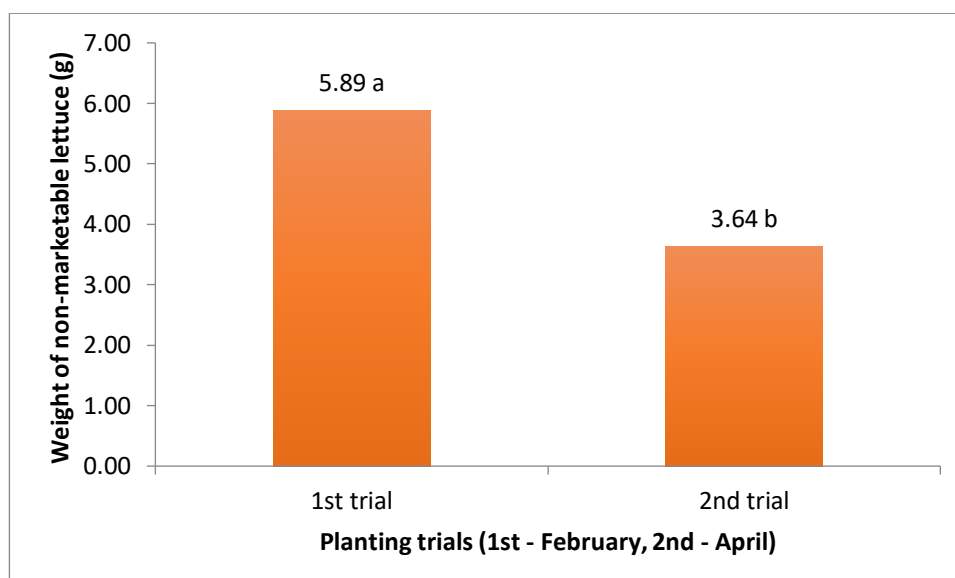


Fig. 9.b. Weight of non-marketable lettuce (g) as affected with the different planting schedules.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 15 of 18

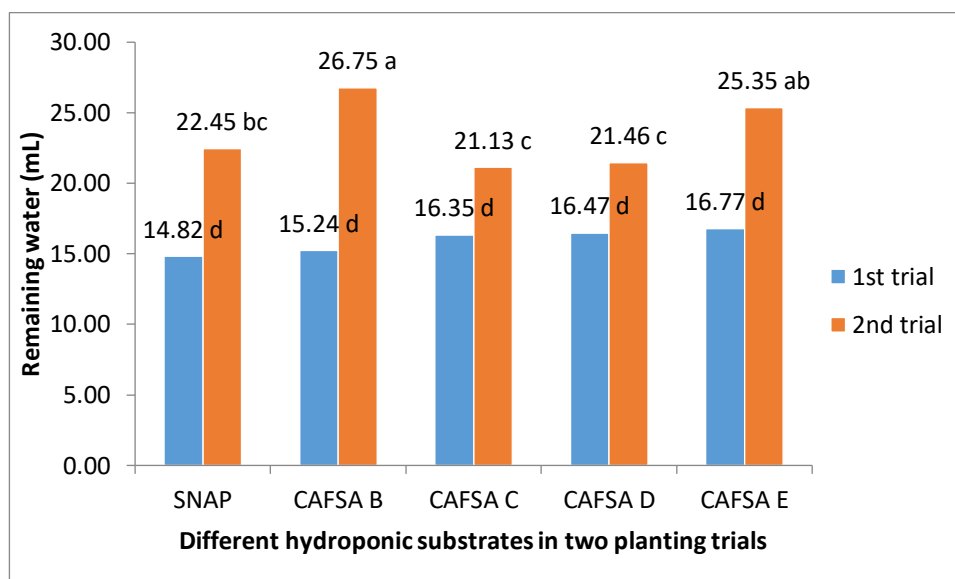


Fig. 10. Volume of the remaining water as affected with the different hydroponic substrates and planting trials.

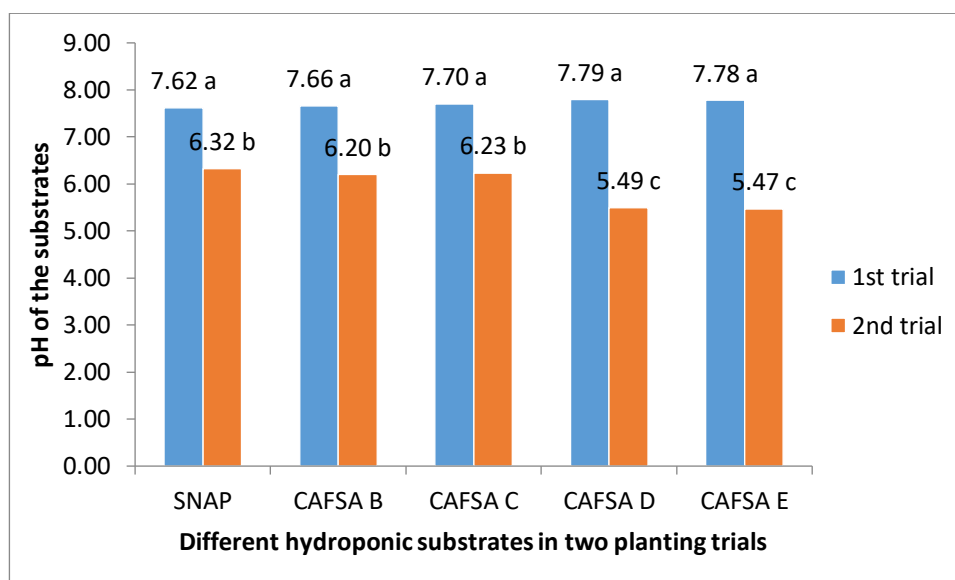


Fig. 11. pH of the water as affected with the different hydroponic substrates and planting trials.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 16 of 18

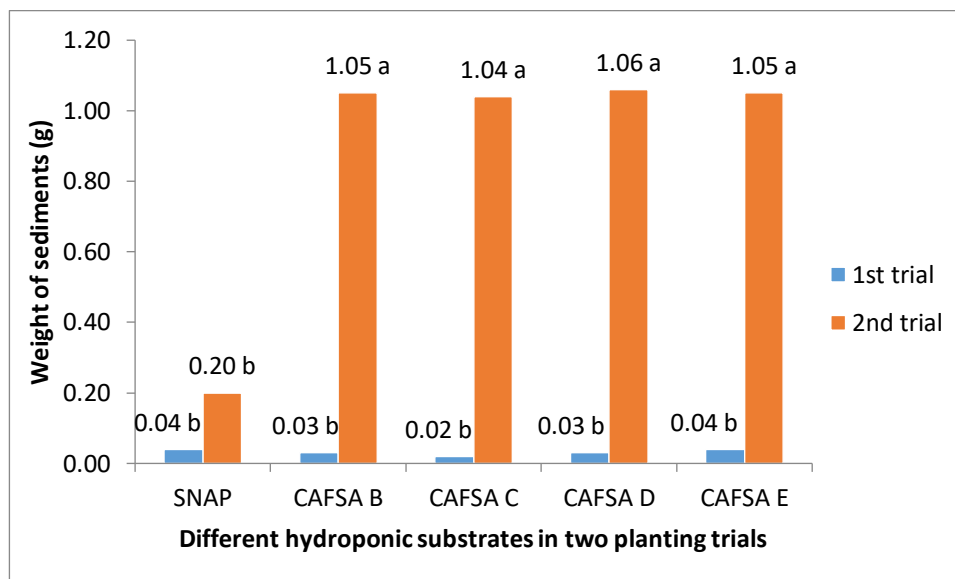


Fig. 12. Weight of sediments (g) as affected with the different hydroponic substrates and planting trials.

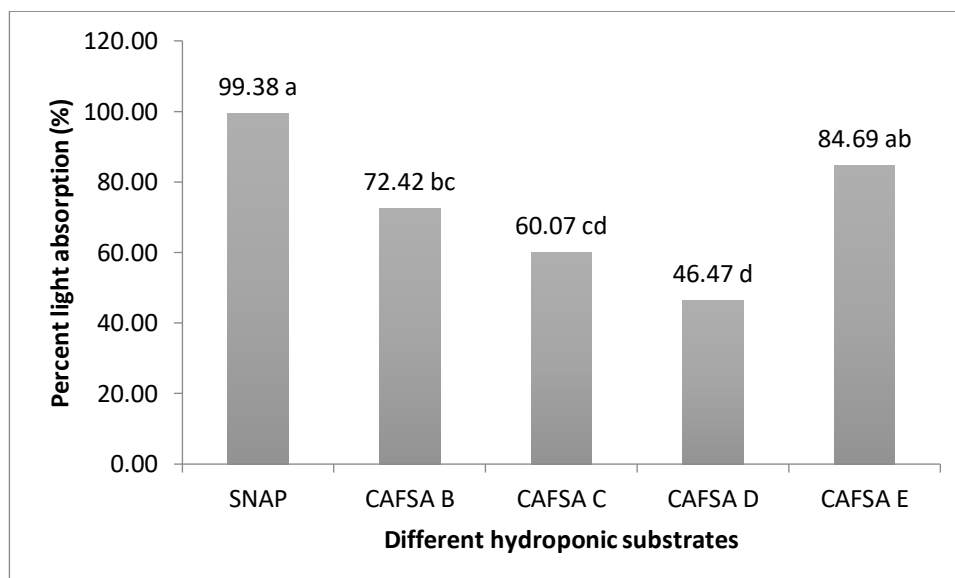


Fig. 14.a. Percent light absorption (%) as affected with the different hydroponic substrates.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 17 of 18

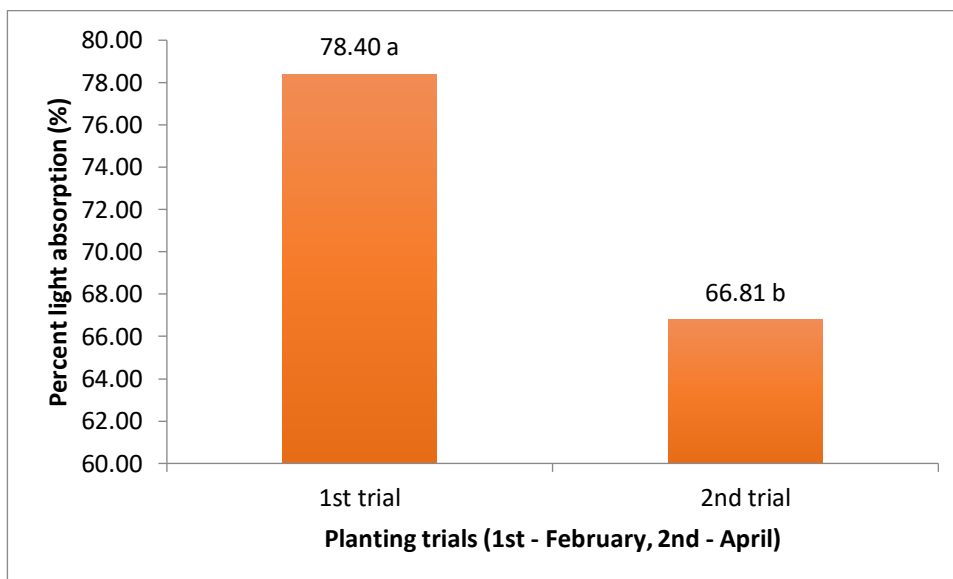


Fig. 14.b. Percent light absorption (%) as affected with the planting trials.



Fig. 15. Color wheel palette.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 18 of 18

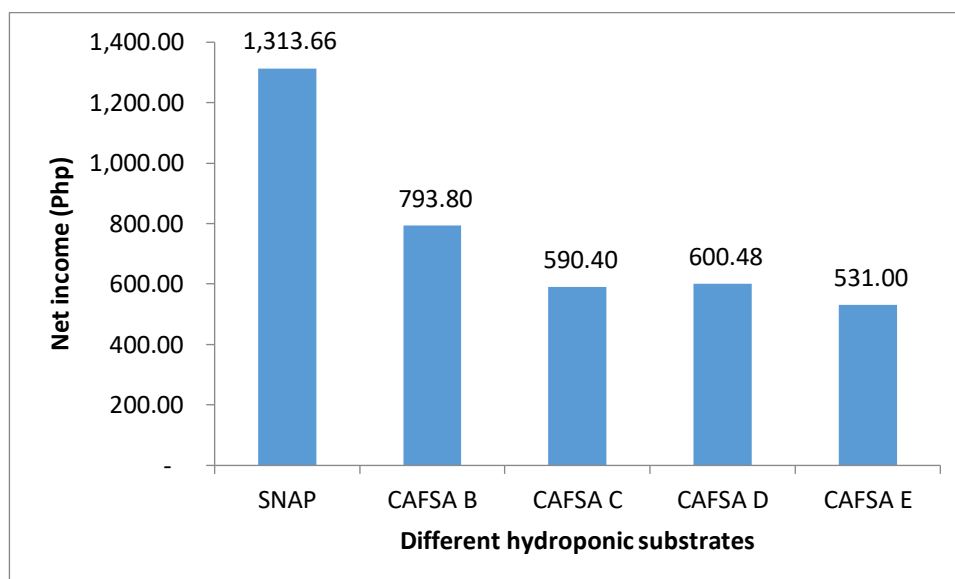


Fig. 16. Net income from the lettuce applied with the different hydroponic substrates in two planting trials.

Table 1. Income over substrate cost of organically produced lettuce under hydroponic cultivation.

PARTICULARS	TREATMENTS				
	SNAP	CAFSA with chicken dung and molasses	CAFSA with cow manure and without molasses,	CAFSA with chicken dung and without molasses	CAFSA with hog manure and without molasses
Cost of Seedlings (P) ^a	180.00	180.00	180.00	180.00	180.00
Amount of substrate	200 ml	3.00 kg	3.00 kg	3.00 kg	3.00 kg
Cost of substrate (P) ^b	800.00	36.00	30.00	30.00	30.00
Nylon Net (P) ^c	0	35.00	35.00	35.00	35.00
Total cost of production (P)	980.00	251.00	245.00	245.00	245.00
Gross Income	1,313.66	793.80	600.48	590.40	531.00

a = seedling cost at P 2.00/seedling

b = for SNAP cost is P 4.00/ml; for CAFSA with molasses P 12.00/kg; CAFSA w/o molasses P 10.00/kg

c = nylon net cost is P35.00/m